


```cpp
template <typename T>
T max(T x, T y)
{
    if (x >= y) {
        return x;
    }
    else return y;
}
```

Basic Document Control Template

Example

Document Title	Document No.	Issue Date
EV Charging Station Installation Guide	PC-DOC-005	09/26/20XX

Revision History

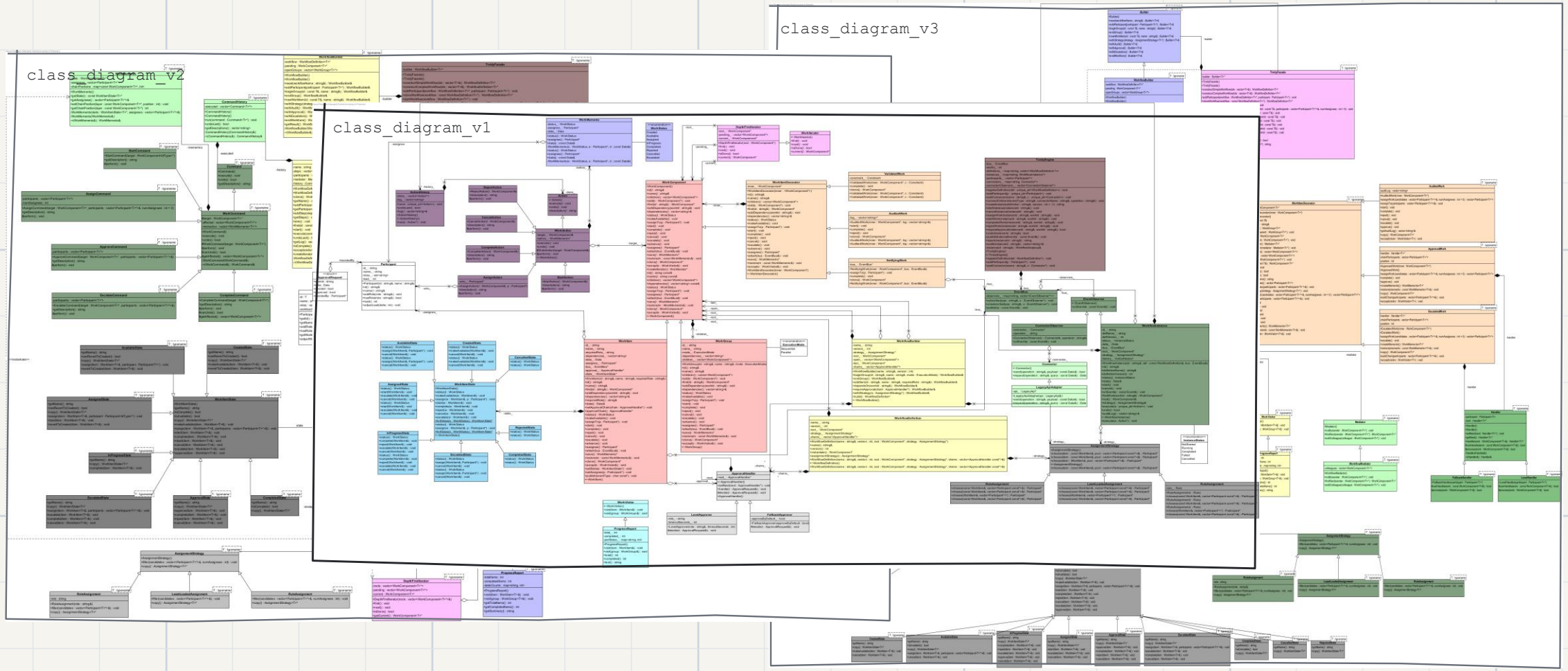
Reason for Revision	Rev. No.	Approved by	Revision Date
Updated connection protocols for faster charging	1.0	Carmen Robertson	07/10/20XX
Added safety procedures for weather conditions	2.0	Jason Desjardins	08/01/20XX
Corrected errors in voltage calibration instructions	3.0	Olivia Carter	08/18/20XX
Expanded guidelines for installation in rural areas	4.0	Sarah Goodwin	09/05/20XX
Included additional maintenance steps	5.0	Victoria Pearson	09/20/20XX

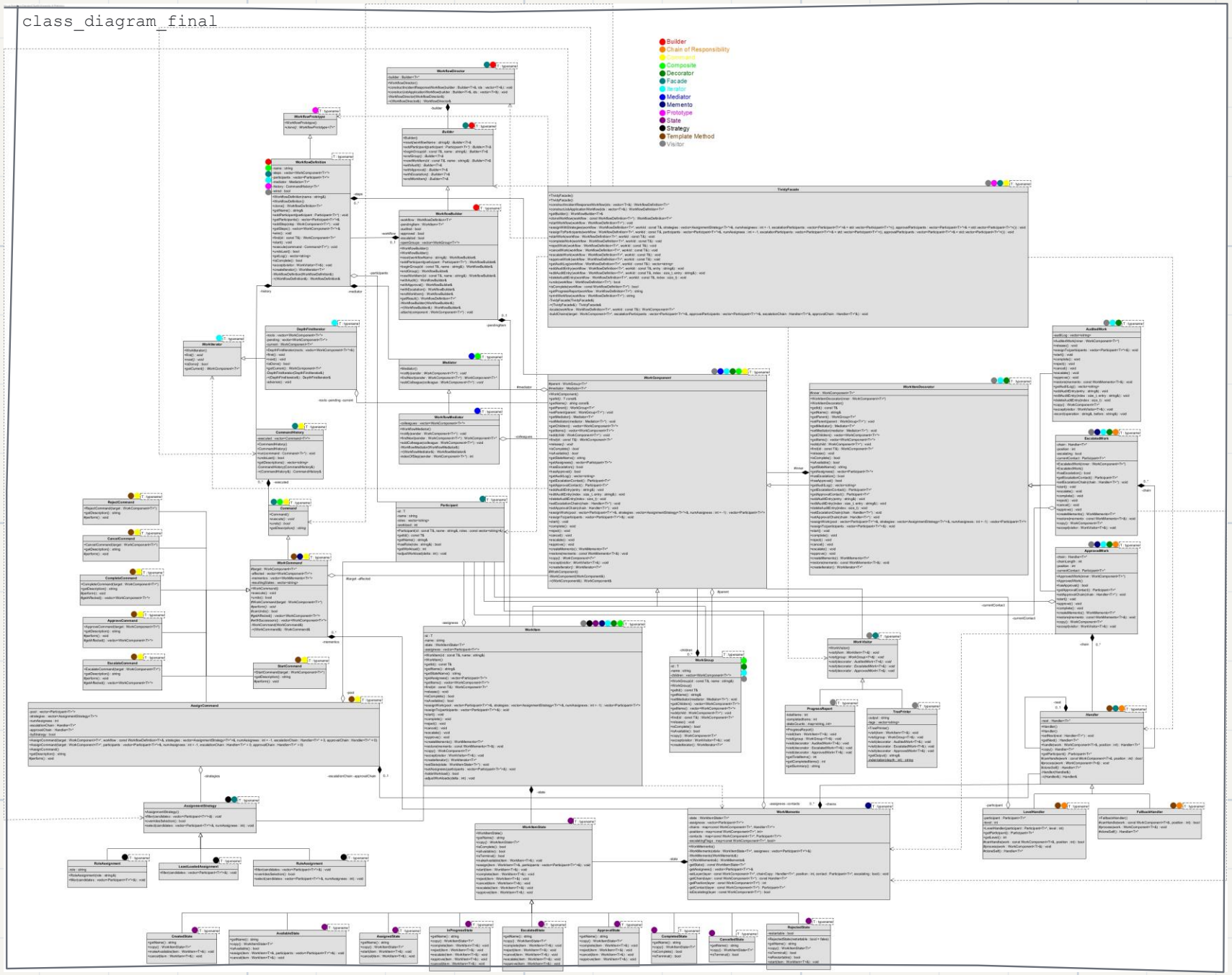
Distribution List

Devon Gomez	Jonathon Wong
Diana Kennedy	Jose Price
Everett Crosse	June Taylor
Fiorella Fitzgerald	Karl Peter
Guadalupe Garcia	Kassandra Hoover

“Go play with it.”

— PROF. MARSHALL




class_diagram_final
Builder
Chain of Responsibility
Command
Composite
Decorator
Facade
Iterator
Mediator
Memento
Prototype
State
Strategy
Template Method
Visitor

TIVIDY

A Generic Workflow Management System

The Power Rangers

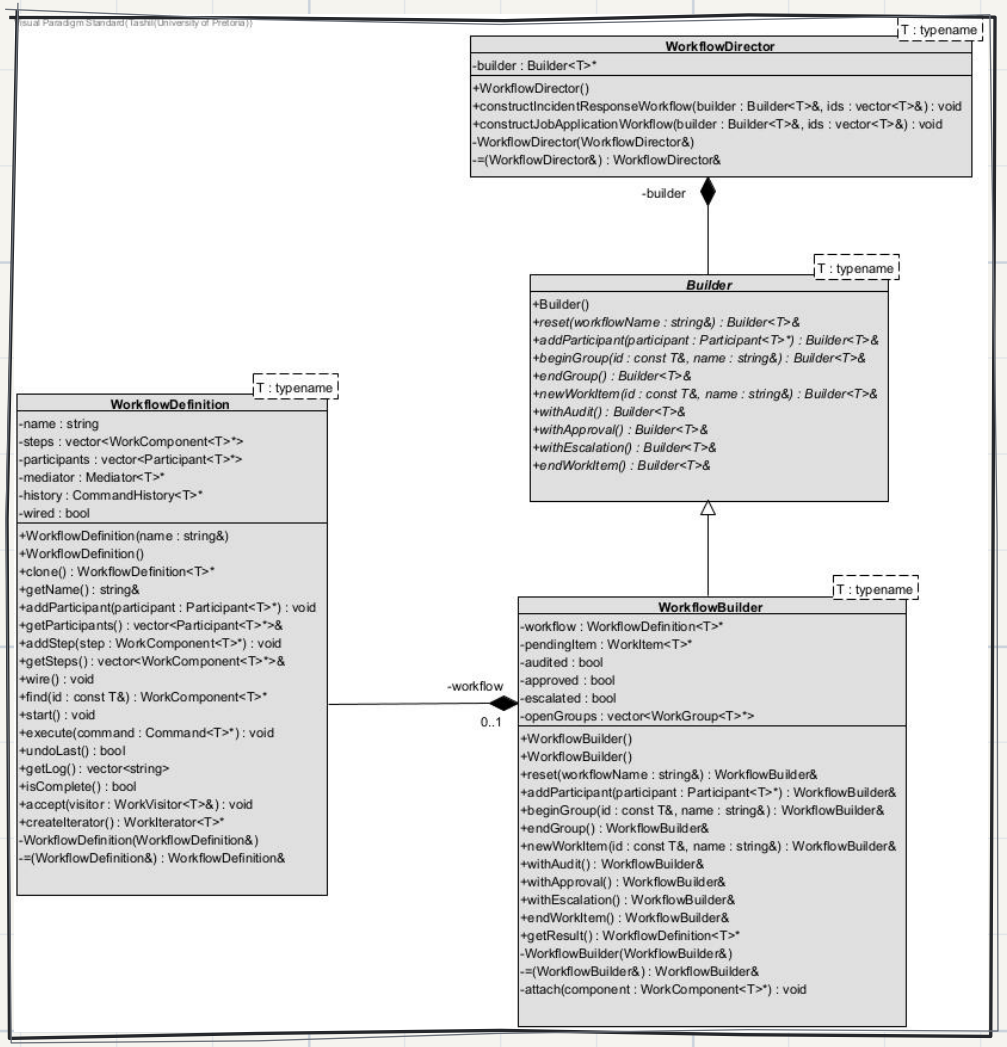


GITHUB REPO QR CODE

PATTERN 01 / 14 — CREATIONAL

Builder

Director tells the Builder which pieces to add, and in what order, to build a complex workflow step by step.

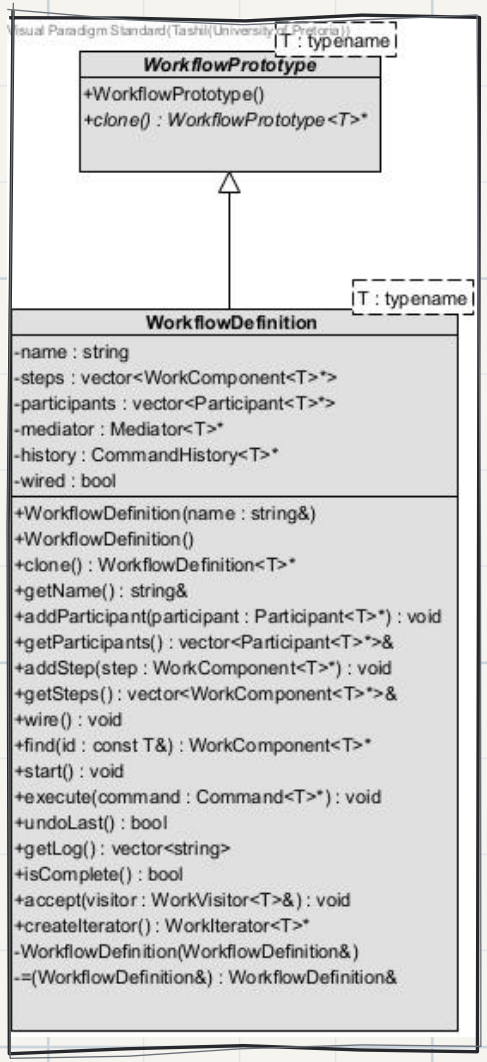


NB: certain participants may have been omitted for visual purposes

PATTERN 02 / 14 — CREATIONAL

Prototype

Clone a finished WorkflowDefinition with a deep copy, rather than rebuilding a complex object each time.

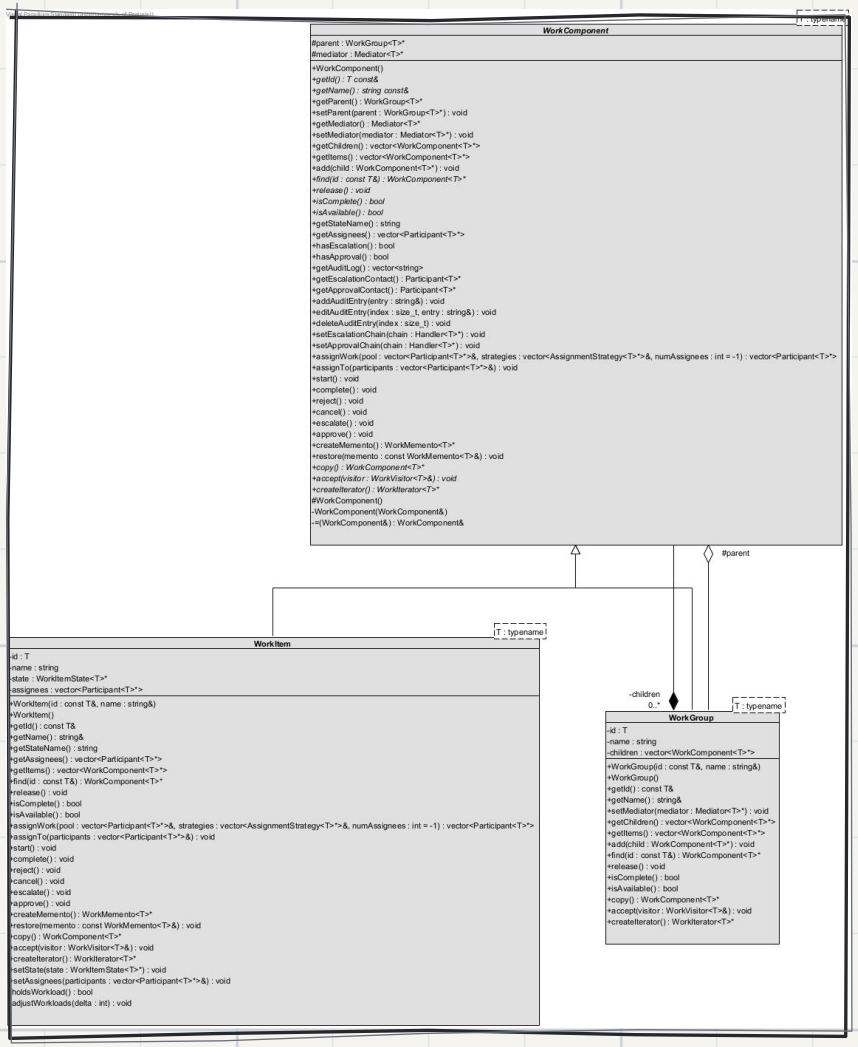


NB: certain participants may have been omitted for visual purposes

PATTERN 03 / 14 — STRUCTURAL

Composite

A WorkItem is one task. A WorkGroup is a step that holds other items or groups

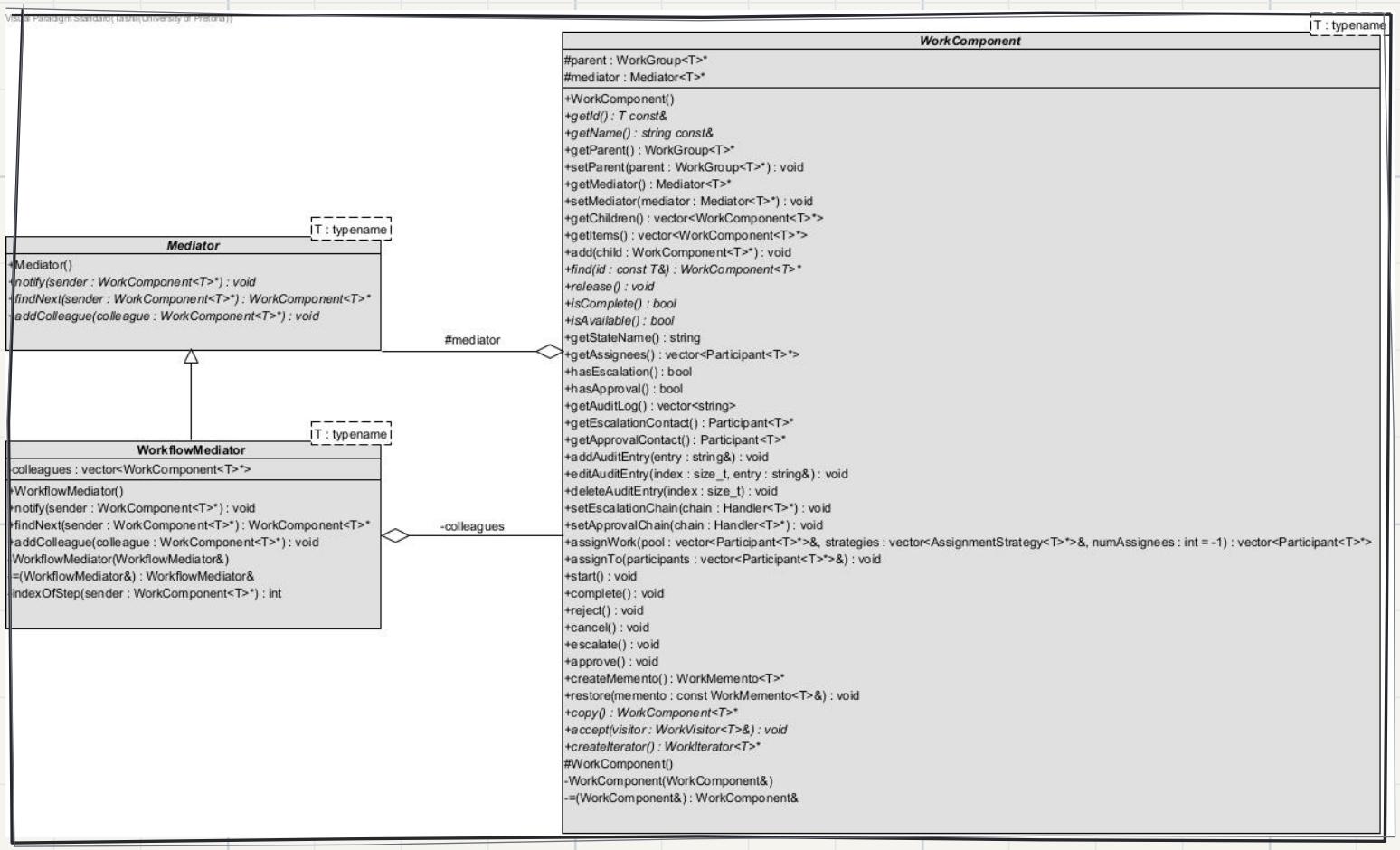


NB: certain participants may have been omitted for visual purposes

PATTERN 04 / 14 — BEHAVIOURAL

Mediator

The Mediator decides when the next component is released.

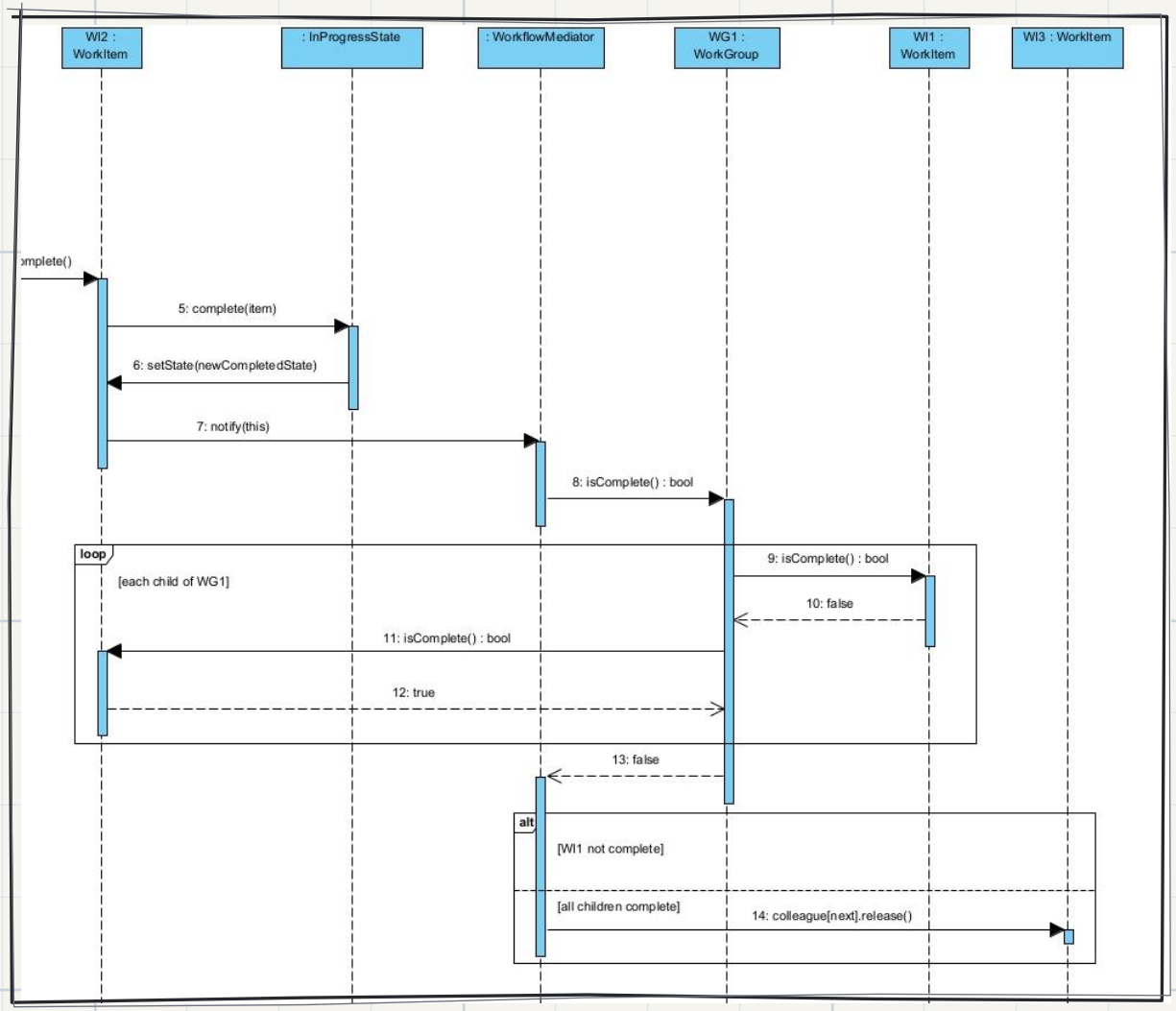


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PATTERN 04 / 14 — BEHAVIOURAL

Mediator

The Mediator decides when the next component is released.

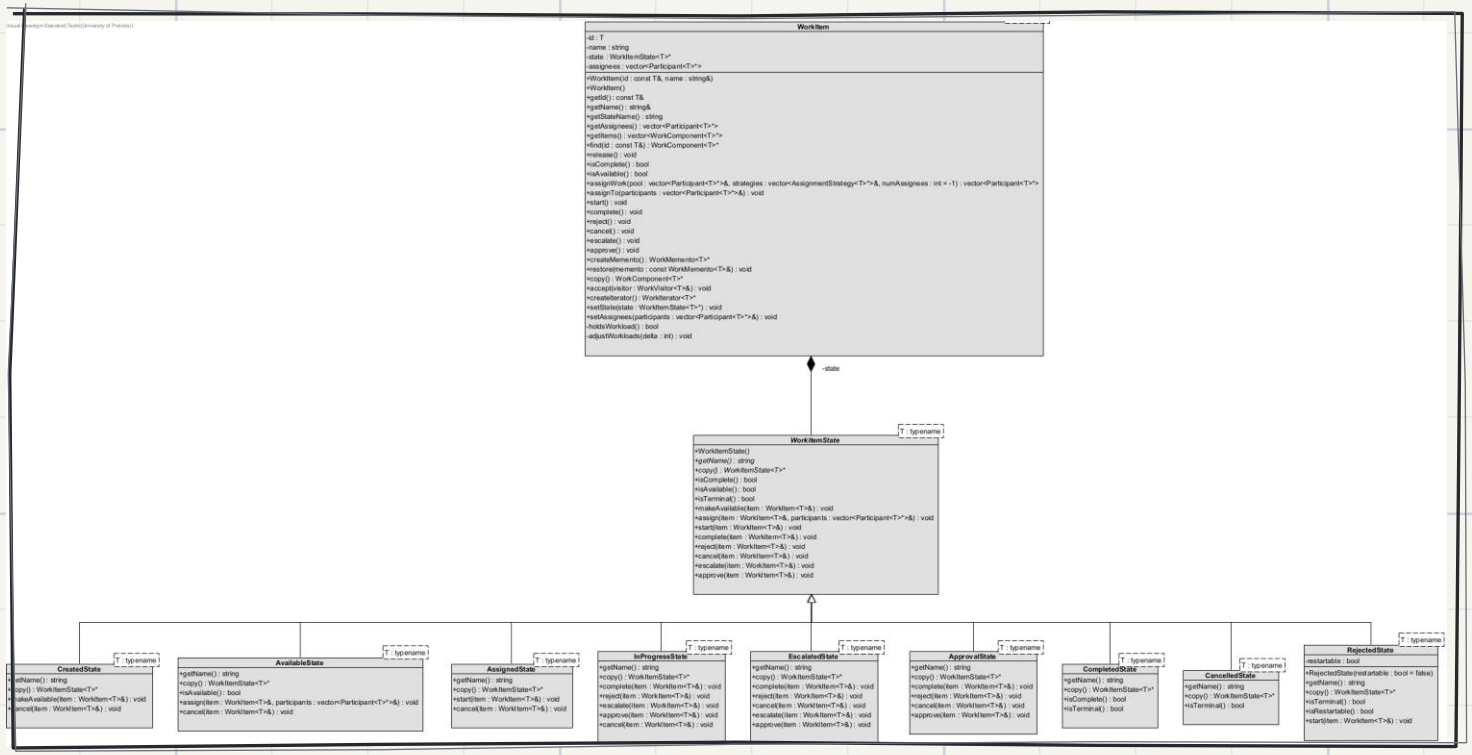


NB: entire diagram may not be shown for visual purposes

PATTERN 05 / 14 — BEHAVIOURAL

State

State decides which operations are legal at any moment.

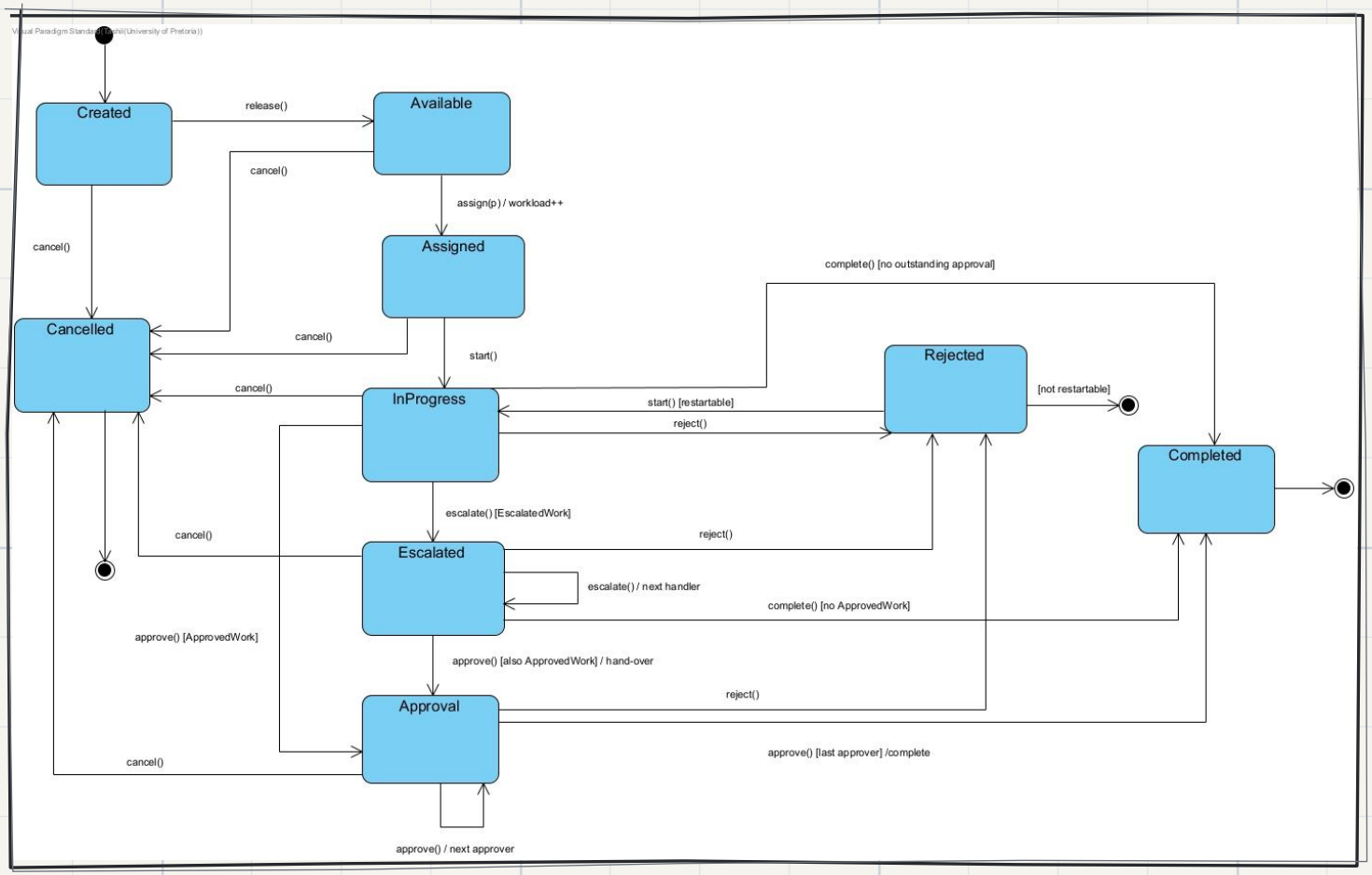


NB: certain participants may have been omitted for visual purposes

PATTERN 05 / 14 — BEHAVIOURAL

State

State decides which operations are legal at any moment.

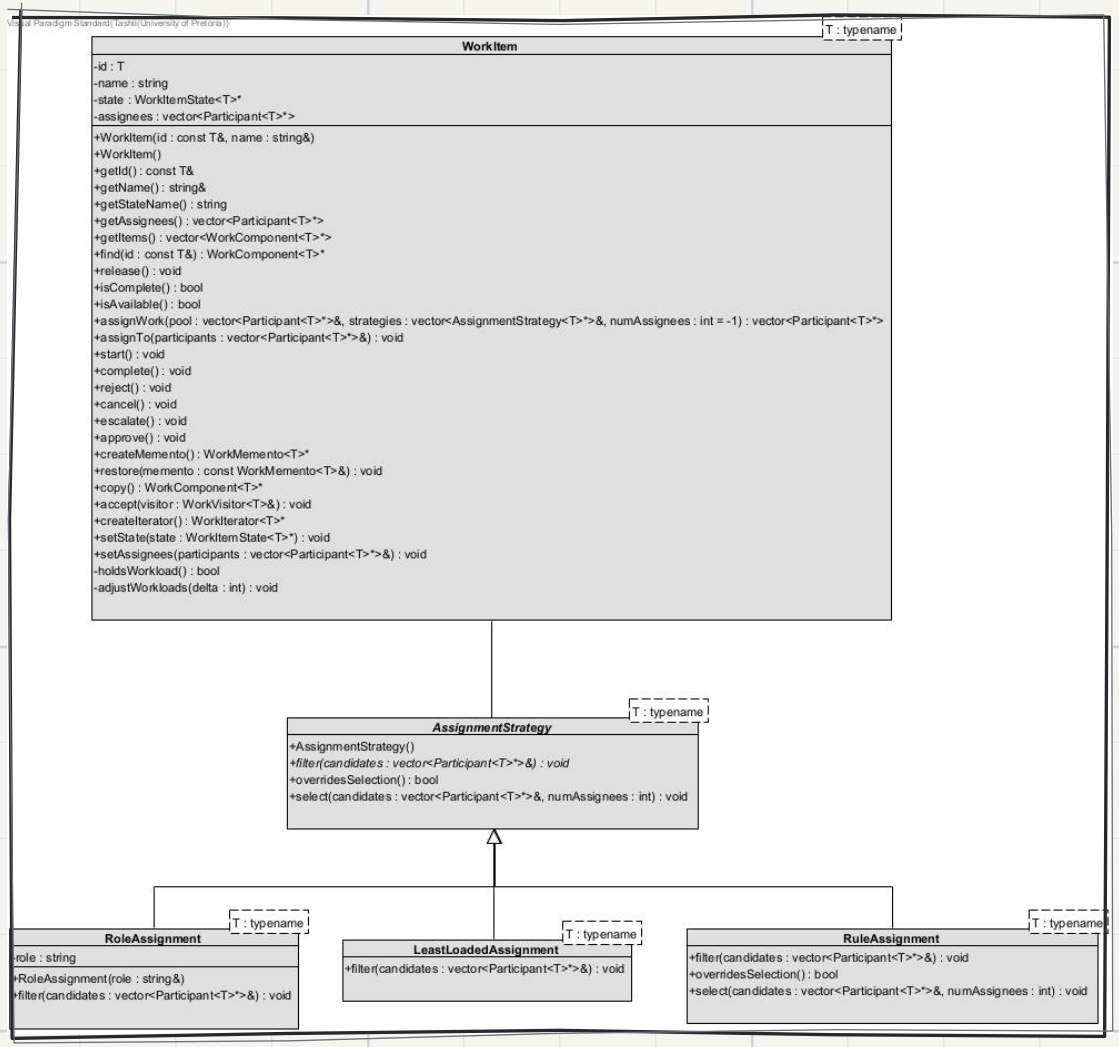


NB: entire diagram may not be shown for visual purposes

PATTERN 06 / 14 — BEHAVIOURAL

Strategy

Assignment algorithms are interchangeable. Each one narrows the pool; with none passed in, assignment is manual.

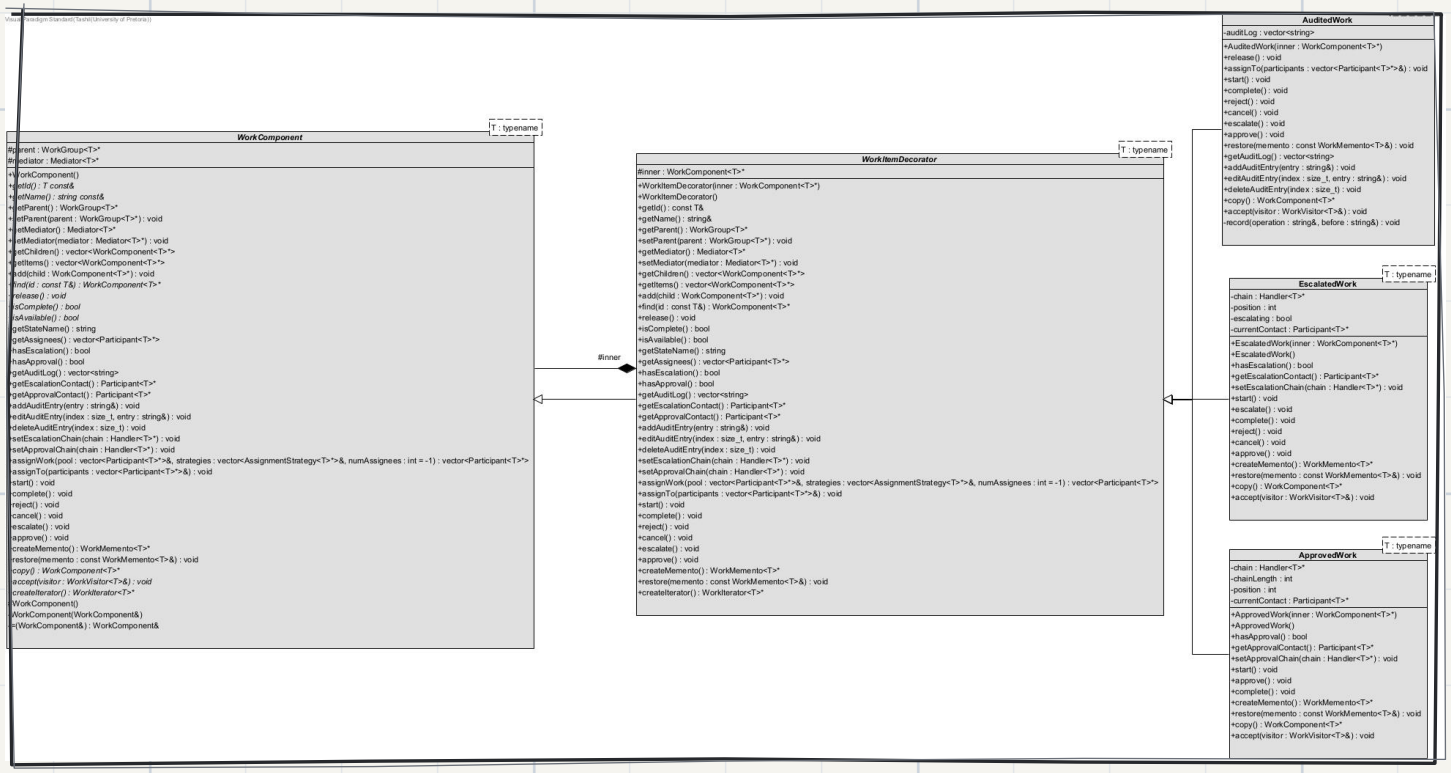


NB: certain participants may have been omitted for visual purposes

PATTERN 07 / 14 — STRUCTURAL

Decorator

Adds extra behaviour such as audit history, escalation or approval, without a subclass for every combination.

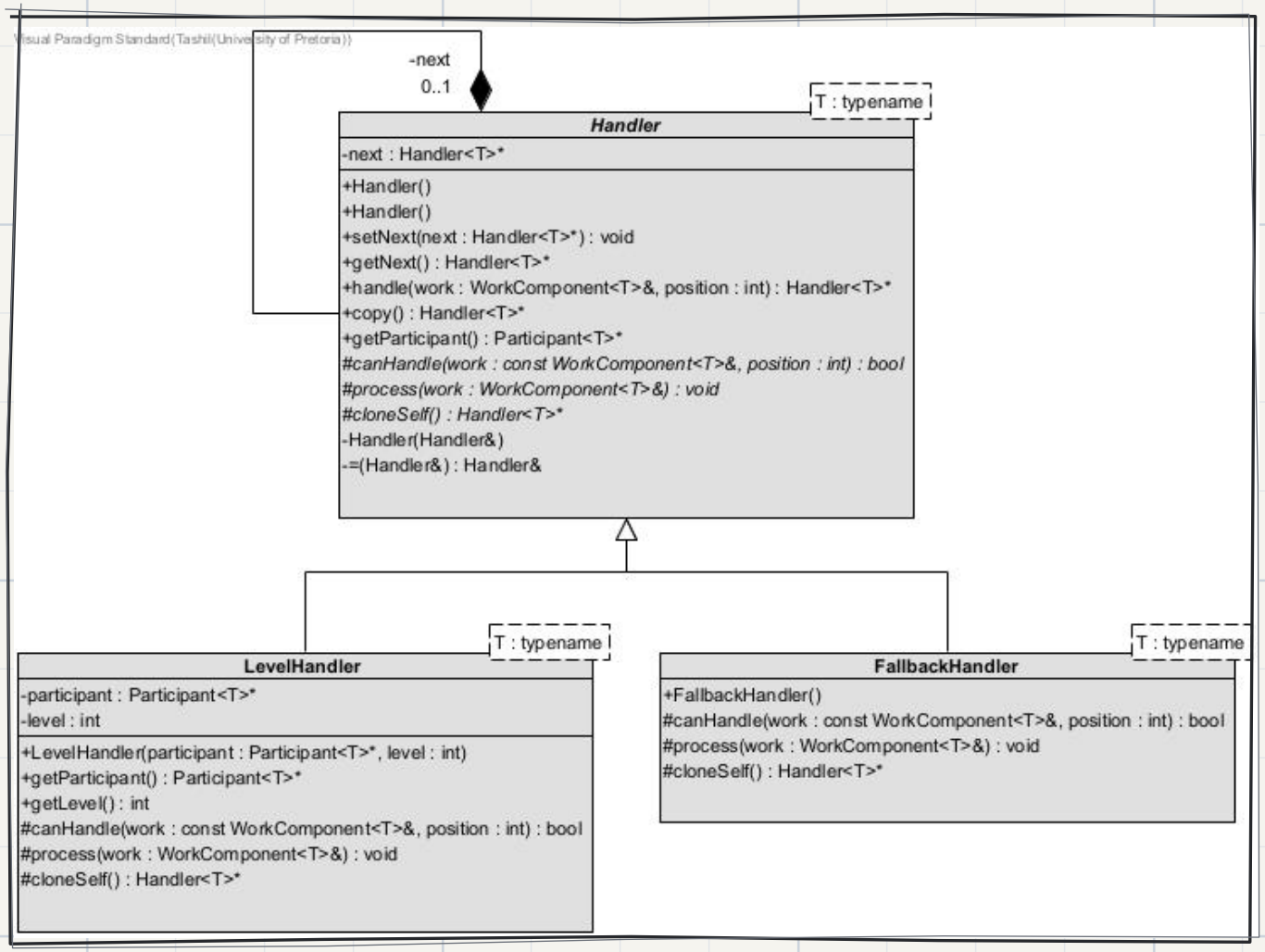


NB: certain participants may have been omitted for visual purposes

PATTERN 08 / 14 — BEHAVIOURAL

Chain of Responsibility

Handlers are linked. A request moves one link at a time until one handles it, and the FallbackHandler always does.

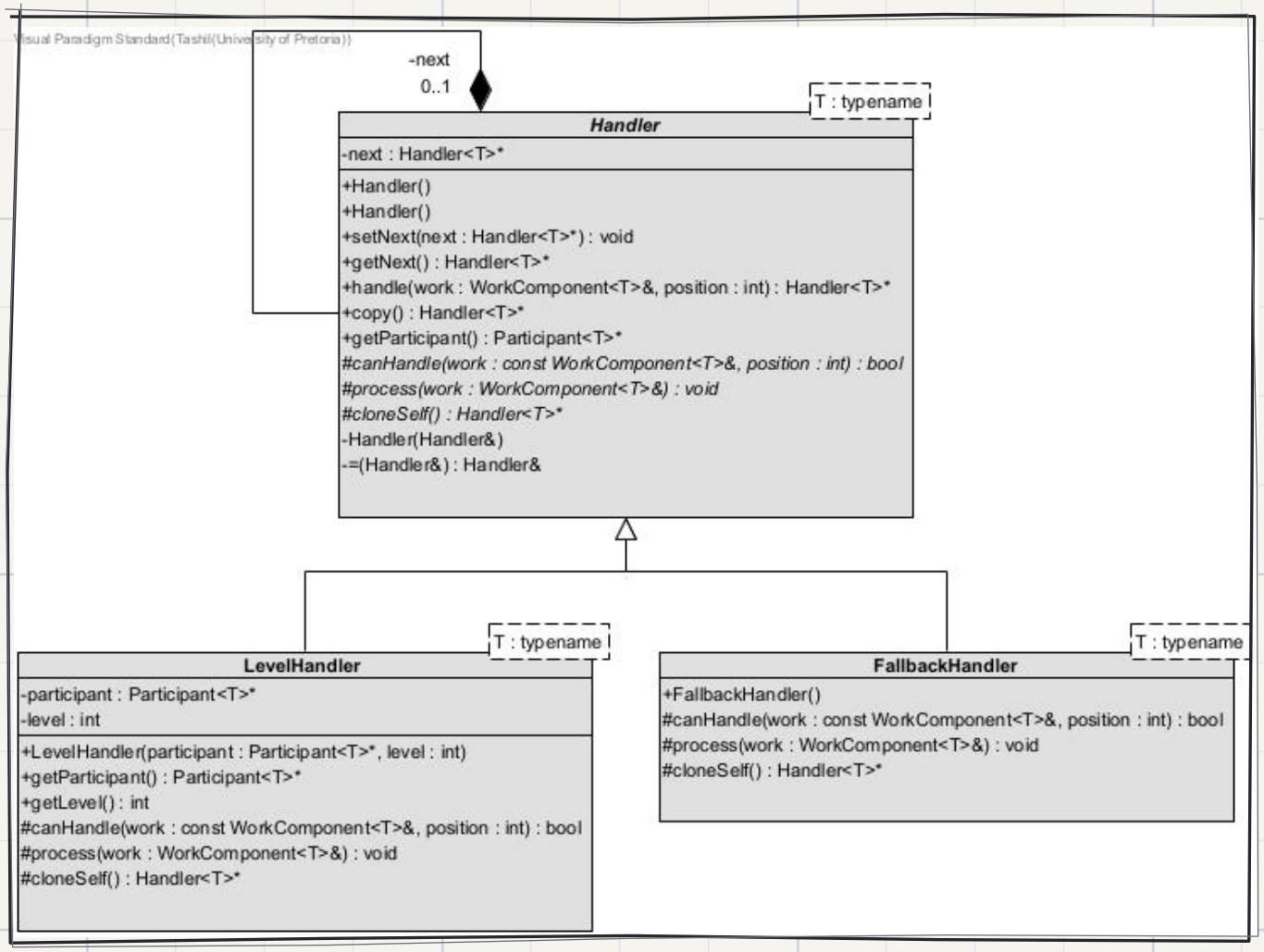


NB: certain participants may have been omitted for visual purposes

PATTERN 09 / 14 — BEHAVIOURAL

Template Method

The skeleton is fixed: check if the handler can process the request, and either run it or pass it along

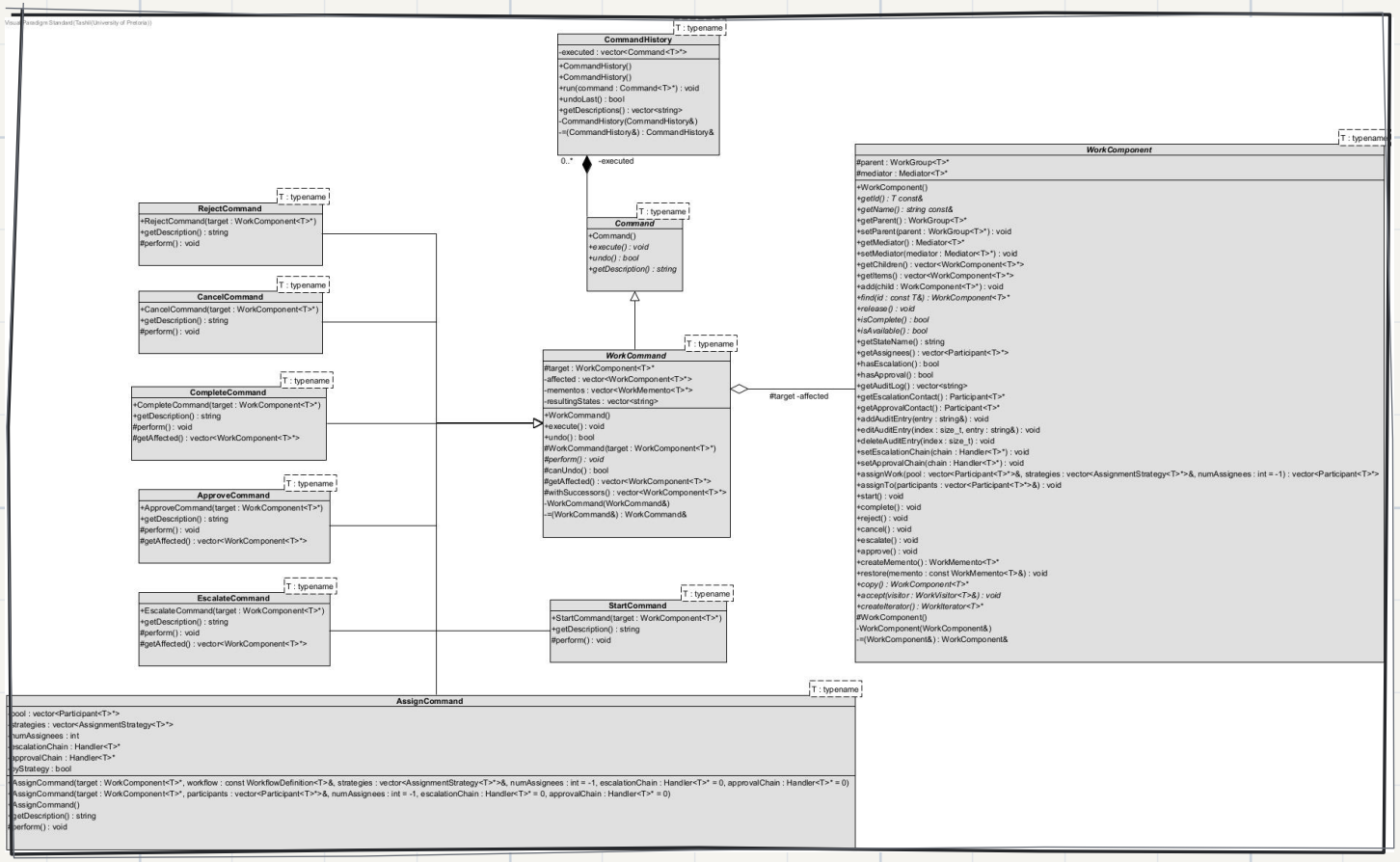


NB: certain participants may have been omitted for visual purposes

PATTERN 10 / 14 — BEHAVIOURAL

Command

Each action is encapsulated as an object, so everything can be tracked and undone.

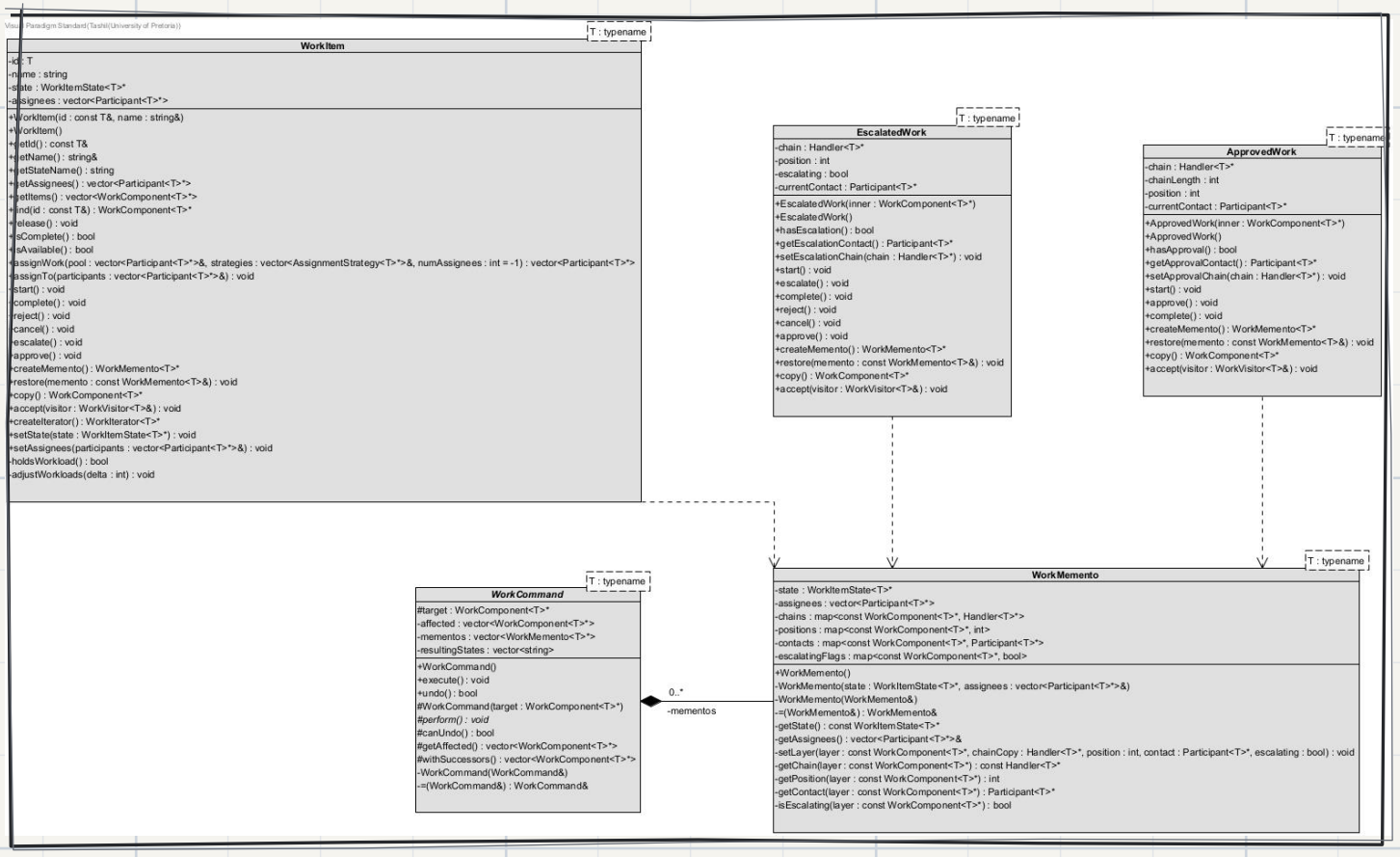


NB: certain participants may have been omitted for visual purposes

PATTERN 11 / 14 — BEHAVIOURAL

Memento

Memento restores the exact previous state without breaking encapsulation. Undo runs newest first.

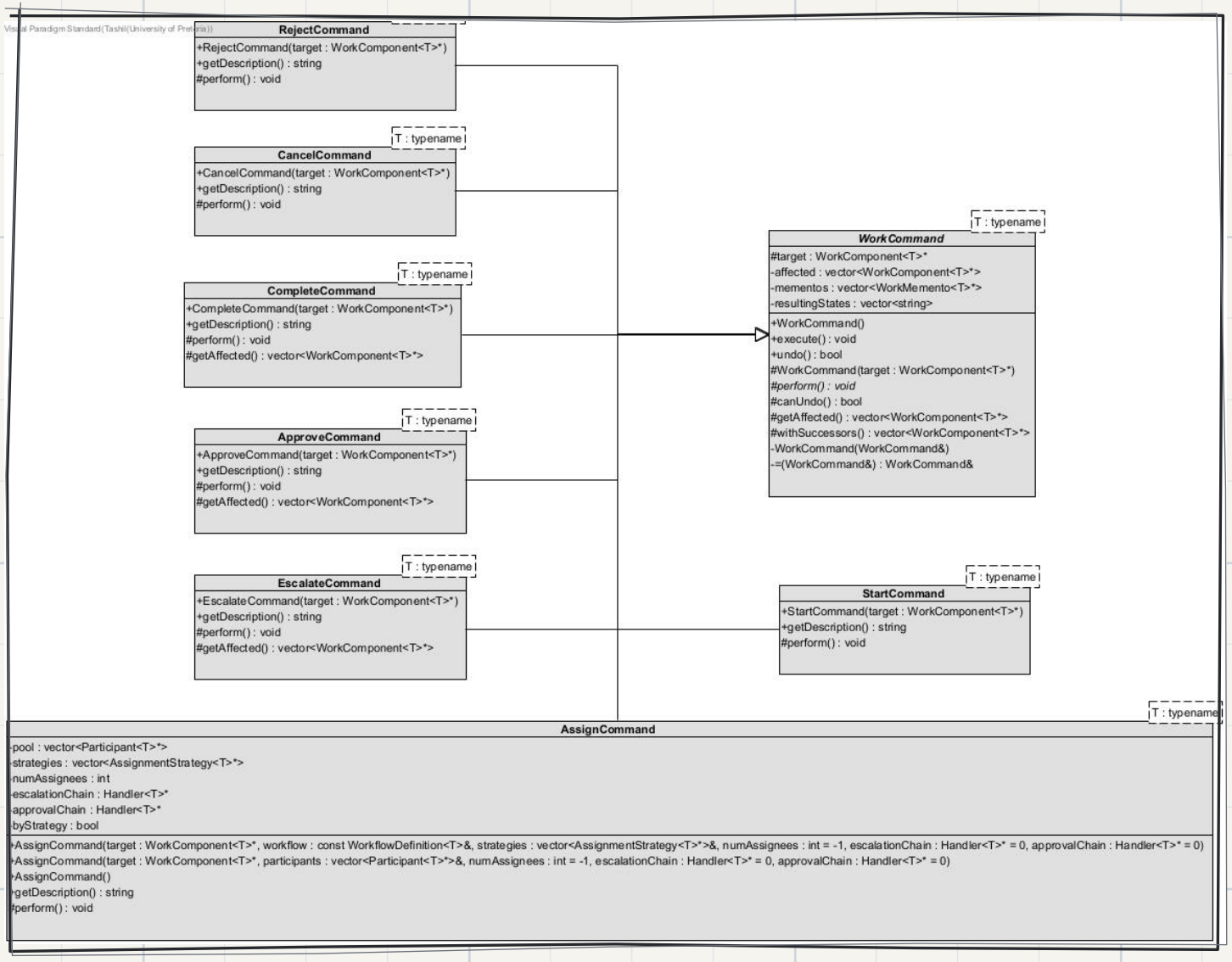


NB: certain participants may have been omitted for visual purposes

PATTERN 09 / 14 — BEHAVIOURAL

Template Method

The skeleton is fixed: snapshot, perform, record. Each ConcreteCommand supplies perform.

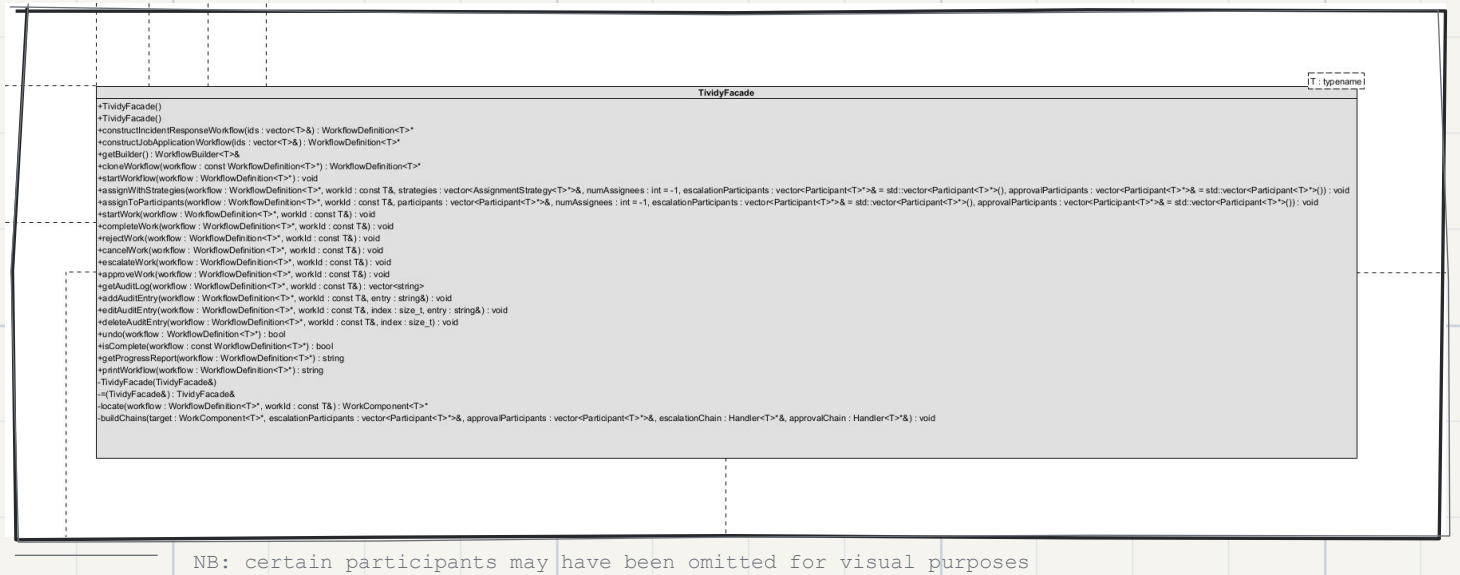


NB: certain participants may have been omitted for visual purposes

PATTERN 12 / 14 — STRUCTURAL

Façade

One unified, high-level interface over the complex subsystem classes, so the client gets a much simpler way in.

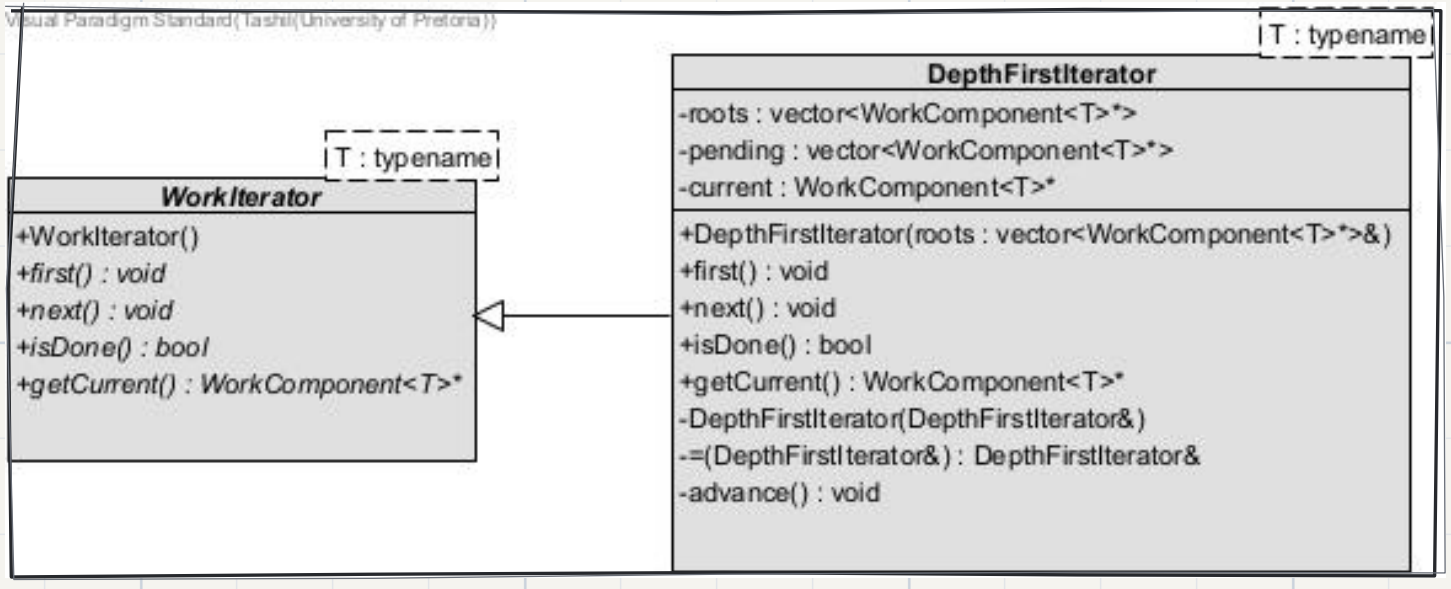


NB: certain participants may have been omitted for visual purposes

PATTERN 13 / 14 — BEHAVIOURAL

Iterator

A DepthFirstIterator walks the workflow tree without exposing how its elements are stored.

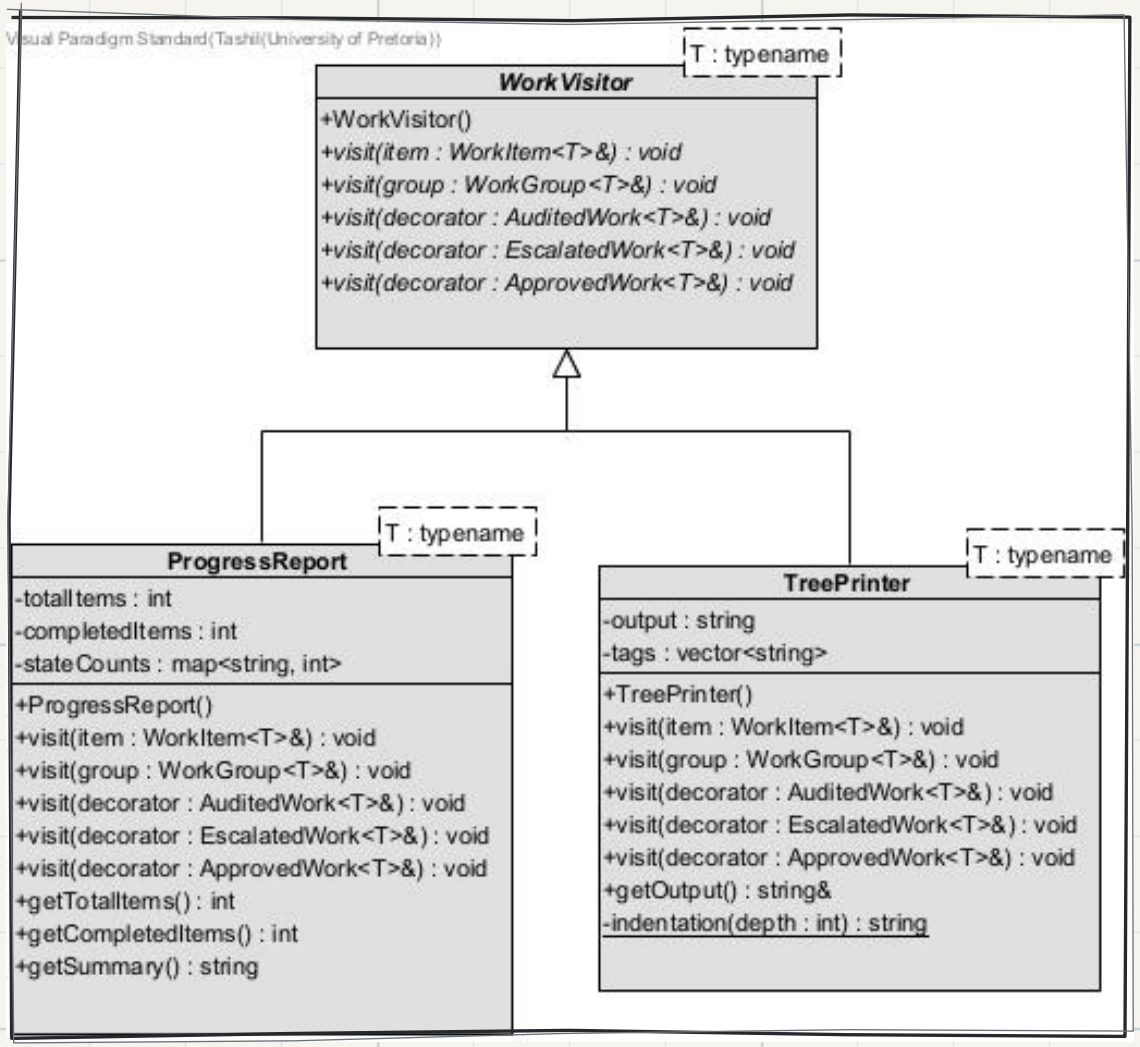


NB: certain participants may have been omitted for visual purposes

PATTERN 14 / 14 — BEHAVIOURAL

Visitor

Reports live in visitors, not in the work classes. ProgressReport counts the states; TreePrinter draws the tree.



NB: certain participants may have been omitted for visual purposes


No Signal

CONCRETE SCENARIO 01 — INCIDENT RESPONSE WORKFLOW

Lights out.

On 29th September 2026, during heavy rain, an explosion at the Waltloo substation caused a 132 kV trip at Njala. A transformer was found faulty, and the City ran three-hour load rotations while it was being repaired.



CONCRETE SCENARIO 02 — JOB APPLICATION WORKFLOW

Tyto Insights

A data consultancy based in Pretoria, founded by our very own University of Pretoria computer science graduates Mark Botros and Inge Odendaal.



01

Initial Screening

Application Review
Reference Check

02

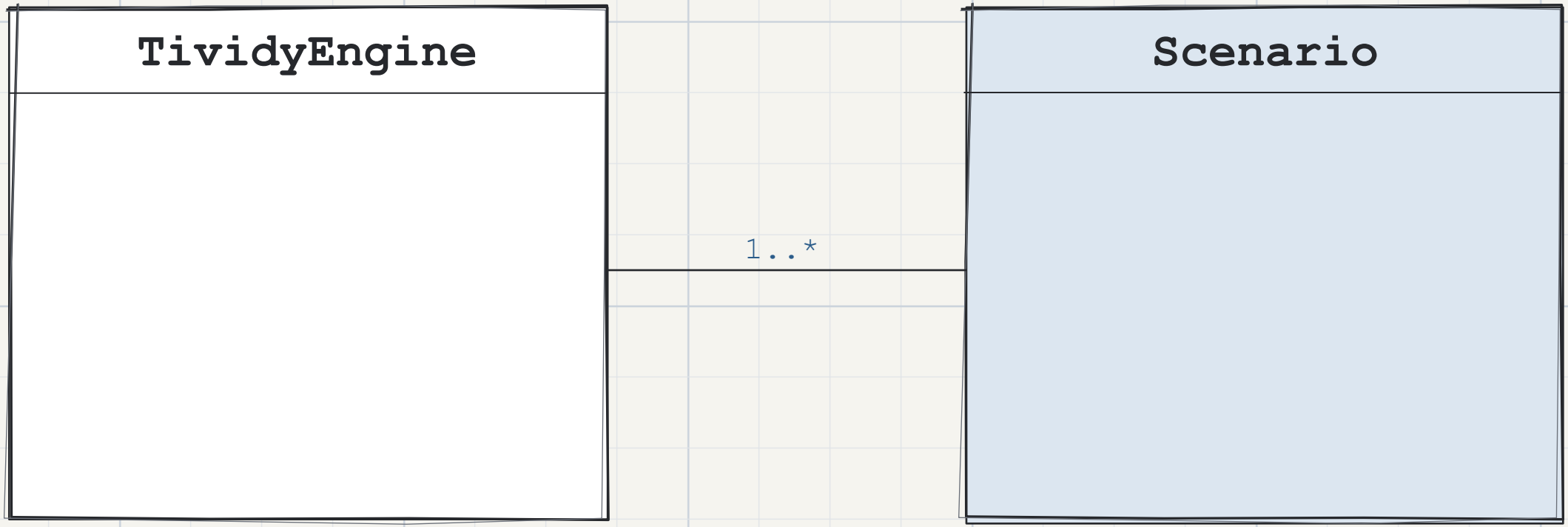
Interviews

Behavioural Interview
Technical Interview

03

Final Decision

Approved?
Rejected?


TividyEngine
1..*
Scenario

Any questions?



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